

■ 3.2.8 Mathematical Constants

I	$i = \sqrt{-1}$
Infinity	∞
Pi	$\pi \simeq 3.14159$
Degree	$\pi/180$: degrees to radians conversion factor
GoldenRatio	$\phi = (1 + \sqrt{5})/2 \simeq 1.61803$
E	$e \simeq 2.71828$
EulerGamma	Euler's constant $\gamma \simeq 0.577216$
Catalan	Catalan's constant $\simeq 0.915966$

Mathematical constants.

Euler's constant `EulerGamma` is given by the limit $\gamma = \lim_{m \rightarrow \infty} (\sum_{k=1}^m \frac{1}{k} - \log m)$. It appears in many integrals, and asymptotic formulae. It is sometimes known as the **Euler-Mascheroni constant**, and denoted C .

Catalan's constant `Catalan` is given by the sum $\sum_{k=0}^{\infty} (-1)^k (2k+1)^{-2}$. It often appears in asymptotic estimates of combinatorial functions.

Mathematical constants can be evaluated to arbitrary precision.

```
In[1]:= N[EulerGamma, 40]
Out[1]= 0.5772156649015328606065120900824024310422
```